Question			Answer	Marks	AO	Guidance
8	(a)		When $x = 0$ number of cakes is 190	**B1** **[1]**	**3.4**	Allow www
8	(b)		(1.20, 50) gives $50 = 1.2m + c$ (2.40, 140) gives $140 = 2.4m + c$ $m = 75,\ c = -40$	**M1** **A1**	**3.3** **3.3**	Setting up simultaneous equations for *m* and *c* Allow for values given or $y = 75x - 40$ seen
			Alternative method $m = \frac{140 - 50}{2.4 - 1.2}$ $m = 75,\ c = -40$	**M1** **A1**		using data to calculate *m* Allow for values given or $y = 75x - 40$ seen
				[2]		
8	(c)		[When *x* is small,] *y* is negative and number of cakes donated cannot be negative	**E1** **[1]**	**3.5b**	Negative *y*-values and *y* cannot be negative must both be stated or implied
8	(d)		Upper bound for demand to exceed supply Lower bound for *y* positive $190 - 70x = 75x - 40$ So $x < \frac{46}{29}$ $y = 75x - 40 > 0$ $x > \frac{40}{75}$	**M1** **A1** **A1** **[3]**		Attempt to find one of the bounds for *x* using their $y = 75x - 40$ Accept $x < 1.586$ or $x < 1.59$ or $x < 1.58$ or $x \leq 1.56$ Allow use of $<$ or $\leq$ Allow for a separate inequality or as part of a combined inequality with lower bound Accept $0.533 < x$ or $0.53 < x$ or $0.54 < x$ or $0.56 \leq x$ Allow use of $<$ or $\leq$ Allow for a separate inequality or as part of a combined inequality with upper bound Note the prices £1.58 and £1.57 do not lead to integer values for the number of cakes, so the fully correct answer is $x \leq 1.56$ Note the prices £0.54 and £0.55 do not lead to integer values for the number of cakes, so the fully correct answer is $0.56 \leq x$